CAPSTONE PROJECT PROPOSAL

Title:

Laluan, An AI-powered Career Roadmap App

By Group 13

A. General Information

Project Title:	**An Ai-Powered Career Roadmap App**

Points of Contact

List the principal individuals who may be contacted for information regarding the project.

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Chapter 1 Introduction/Proposal

1.1. Executive Summary

In two or three paragraphs, provide a brief overview of this project and the contents of this document.

Current modes for career guidance are very limited in their ability to help students. Existing ones use pre-existing questionnaires and a one-size fit all recommendation model that fails to capture a student's evolving interests. These systems often fail in reality due to the rapidly evolving jobs in a highly competitive market. These systems also tend to overlook industry trends, skill demands and personalized career development paths, thus students are provided with a generalized career path that lacks personality and practical guidance.

Thus, Project Laluan is created in order to address the issues of career guidance. Laluan is an android application that leverages the benefits of AI in order to help students to find careers that are suited for them. This is achieved through examining a student's interests, skills and talents before putting the information through an LLM to generate a personalized list of skills and development dashboard for users to nurture. This project is created to address the growing issues of secure employment and career uncertainty after graduation by providing a centralized and interactive platform that guides users towards a suitable career based on the target's strengths, interests, weaknesses and aspirations. The project also provides a calendar system to provide a scheduling benefit for students to find workshops and career fairs.

1.2. Project Purpose

In two or three paragraphs, provide a brief overview of this project and the contents of this document.

1.2.1. Problem Statement

The problem statement should clearly describe a relevant issue, supported by citations, and justify how it can be addressed using computer science or software engineering solutions. It must highlight the problem context, its significance, and the proposed computing-based approach.

There has been a trend of crisis for final-year students when it comes to choosing a clear career direction. Personalised help is often hard to come by and there is so much information out there that it ends up feeling overwhelmed . The advice they get is often fuzzy, they don't see much of the real working world. University career offices are overwhelmed too, with way more students than advisors can handle, so true one-on-one support feels out of reach for most. On top of that, the traditional counseling methods just can't keep pace with fresh job market updates, sticking students with information that is often outdated and less useful for current decisions (F1000Research, 2026).

At the same time, rapid technological advancement continues to reshape the global workforce, increasing the importance of self-awareness, adaptability, and continuous upskilling. However, many students still struggle to evaluate their current competencies against evolving industry standards, often leading to career indecision, lack of confidence, and poor career planning

We need a scalable, tech-savvy fix for this. Human counselors alone can't realistically handle large student populations. That's where Laluan comes in, a software engineering-based solution that integrates Generative AI with Large Language Models (LLMs) to provide more personalized and adaptive career guidance. It processes large amounts of industry and skills data in real time, helping students identify career paths that better match their strengths, interests, and current market needs. Studies have shown that AI-driven career guidance systems that has dynamic skill mapping and adaptive mentoring can reduce career anxiety while improving recommendation quality (F1000Research, 2026).

Laluan also avoids users becoming too dependent on AI suggestions. It was found that excessive reliance on Generative AI may reduce users' cognitive effort and critical thinking engagement (Lee et al., 2025). To address this, Laluan includes an interactive Guidance Loop dashboard that keeps students actively involved through reflection, feedback, and continuous decision-making. This ensures students are not just accepting AI recommendations blindly, but instead actively shaping their own career direction.

1.2.2. Project Objectives

Each objective must begin with an appropriate action verb (e.g., design, develop, implement, evaluate, analyze), be measurable and achievable within the project scope, and align directly with the problem statements, clearly demonstrating how it contributes to addressing the identified issues.

Project Objectives:

- To develop an AI-powered career guidance chatbot using Large Language Models (LLMs) that analyzes student profiles, provide personalized career roadmaps, and automate mentor matching to help students make clearer career decisions.
- To implement a Skill Gap Visualizer that compares a student's current skills with real-time industry requirements and recommends suitable learning resources to help them improve and stay industry-ready.
- To design an interactive web-based interface featuring an automated "Guidance Loop" for continuous progress tracking, alongside a centralized Campus Calendar for university event synchronization.
- To integrate a secure digital portfolio known as the Verified Skills Passport by applying cryptographic hashing techniques to ensure certificates, achievements, and skill records remain authentic.
- To evaluate the usability, functionality, and effectiveness of the Laluan application through User Acceptance Testing (UAT) and user feedback collection to ensure the system meets student needs and expectations.

1.2.3. Functionalities

No.	Proposed Functionality	Problems Solved/ Opportunities
1	Student Career Profile Setup	Allows students to enter their education background, skills, interests, strengths, weaknesses, and career goals so the system can understand them better.
2	AI Career Matching	Helps students find career options that match their profile instead of giving the same advice to everyone.
3	Career Roadmap Generator	Creates a clear step-by-step plan for students to follow, such as what skills to learn and what actions to take next.

4	Skill Gap Visualizer	Shows students what skills they already have and what skills they still need for their chosen career.
5	Guidance Loop Dashboard	Lets students review the AI suggestions, give feedback, and update their career direction instead of fully depending on the AI.
6	AI Career Chatbot	Gives students quick answers and guidance when they have questions about career paths, skills, or preparation.
7	Goal setting featureResume Improvement Tool	Helps students improve their resume by giving simple suggestions on content, structure, and missing information.
8	Interview Practice Feature	Allows students to practise common interview questions and improve their confidence before real interviews.
9	Learning Resource Suggestions	Recommends useful courses, workshops, videos, or learning materials based on the student's missing skills.
10	Campus Career Calendar	Shows career fairs, workshops, talks, and university career events in one place so students do not miss useful opportunities.
11	Verified Skills Passport	Keeps students' skills, certificates, and achievements in one section so they can easily show their progress and preparation.
12	Progress and Milestone Tracker	Helps students track completed tasks, learned skills, and roadmap progress over time.

1.3. Assumptions

State any assumptions made in the project. Assumptions are conditions that are accepted as true without proof and are necessary to simplify the problem or define the project scope. These assumptions must be realistic, clearly stated, and justified.

1. The assumptions are that the users (mainly university students) have access to smartphones or computers that have stable internet connections to communicate with the web-based application.
2. The assumption is that the users will be willing to share proper and adequate personal data including academic history, skills, interests and career objectives to facilitate effective AI-based analysis and recommendations.
3. The assumption is that there are enough and relevant sources of data (e.g., career pathways, job market trends, and skill requirements) to support training and operation of the AI models used in the system.
4. It is assumed that generative AI and large language models can generate reasonably accurate, relevant, and personalized career recommendations based on user input.
5.The assumption is that the system will be run in a controlled academic or prototype setting and not full-scale industrial deployment is required at this point.
6. It is presumed that external services or APIs (in case it is integrated, e.g., learning resources or job trend data) exist, are reliable, and do not have a significant impact on system performance.
7.The assumption is that users will actively use the system capabilities like tracking career roadmap, suggesting important skills to develop, and the chatbot to obtain meaningful results.
8.The assumption made is that there should be basic data privacy and security measures in place to cater to this project phase and that no highly sensitive or regulated personal data will be handled.

1.4. Project Description, Scope and Management Milestones

1.4.1Project Description

Describe the project approach, specific solution, customer(s), and benefits.

Laluan is an artificial intelligence-based web application that will assist university students, particularly graduating students, in better planning and developing their career paths. The system leverages generative artificial intelligence and large language models to analyze users' academic background, personal interests, skills, and career goals. According to the analysis, Laluan offers personalized career advice, career development plans, resume tips, and interview simulator via an intelligent chatbot.

The main target clients of this system are university students, especially students who are nearing the graduation period who require assistance in making informed decisions about career choices and preparing skills for future employment opportunities. Based on the analysis of skills and talents performed by a Machine Learning model, Laluan will build a personalized career development roadmap including, skill gaps, resume improvement suggestions, Explainable AI recommendations and an AI chatbot that assist user's through interview preparation.

The project will seek to offer a smarter, more efficient and personalized approach to career planning than is the case with the traditional career counseling approaches. The system will be able to enhance students career readiness, confidence in their communication, and decision making capability by integrating AI technologies.

1.4.2 Scope

The project scope defines all of the products and services provided by a project, and identifies the limits of the project. The project scope addresses the who, what, where, when, and why of a project.

Who?
The Primary Consumer demographic this project aims to provide services to is university students, primarily students nearing graduation period

Where?
The project is developed as an Android-based application and webapp that connects with the Taylor's site accessible through both mobile and static devices, allowing it's systems to be used conveniently within university environment.

Why?
The project aims to address limitations of traditional and static career guidance systems,

which often provided generalized jobs that usually doesn't personalized to a user's talents and interests in a competitive job market. Laluan seeks provide an adaptable, personalized solution towards improving student's career readiness to prepare for future industry opportunities **In Scope** 1.User registration and login system 2.User profile creation and management 3.Analysis of users' academic background, skills, and interests 4.AI-based career recommendation system 5.Personalized career roadmap generation 6.Resume analysis and improvement suggestions 7.AI chatbot for interview simulation and communication practice 8.User data storage and profile management 9.Campus Event Calendar 10. Skill Gap Dashboard 11. Security and Authentication Feature **Out of Scope** 1.Real-time job application submission 2.Direct recruitment services with companies 3.Integration with external HR or recruitment platforms 4.Paid career consulting or coaching services 5.Mobile application development in the current project phase 6. Blockchain based certification system 7. Paid Career consultation or Recruitment platforms

1.4.3 Summary of Milestones and Deliverables

Provide a list of milestones and deliverables with estimated date and duration.

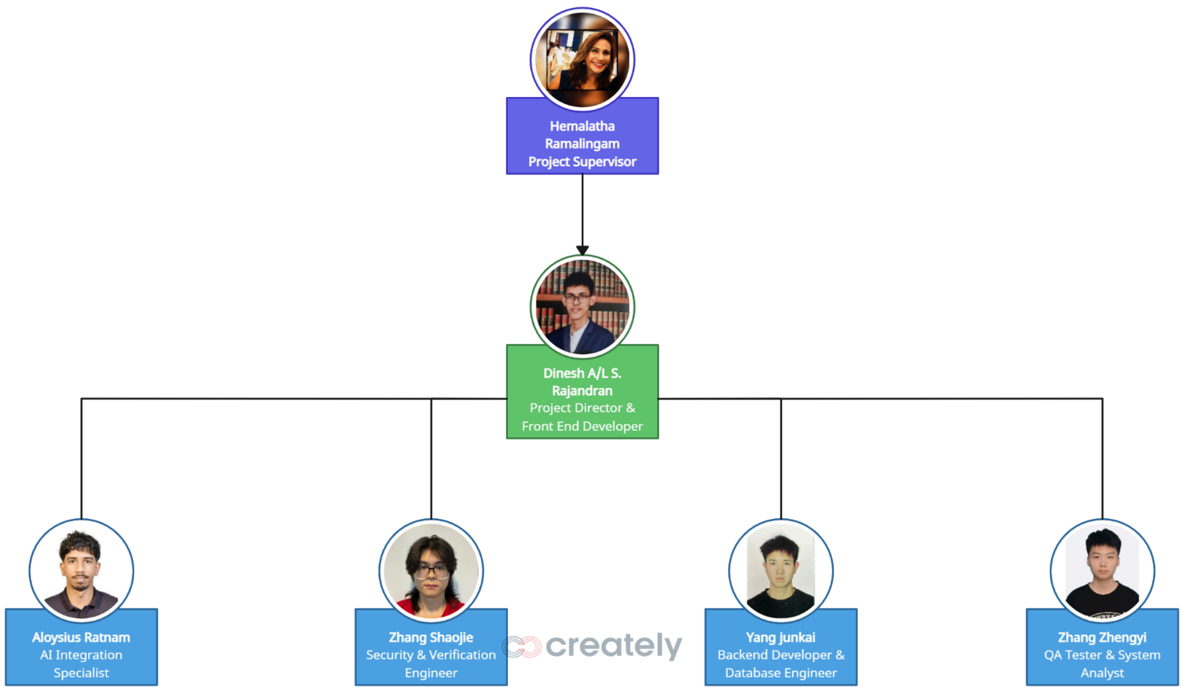
Milestone No.	Milestone / Task	Person Responsible	Expected Duration
Phase 1	**PROPOSAL & PLANNING STAGE**		
001	Client requirements & problem statement finalization	Dinesh	3 days
002	Literature review on AI algorithms and LLMs	Aloysius	4 days
003	Functional requirements & system scope preparation	Zhang Zhengyi	4 days

Mileston e No.	Milestone / Task	Person Responsible	Expected Duration
004	Final proposal document compilation and submission	All Members	3 days
Phase 2	**SYSTEM DESIGN STAGE**		
005	UI/UX Wireframing and Figma Prototype design	Dinesh	7 days
006	Database schema design and System Architecture	Yang Junkai	6 days
007	AI Prompt Engineering & API Logic Planning	Aloysius	5 days
008	Cryptographic hashing & security protocol planning	Zhang Shaojie	4 days
Phase 3	**PROTOTYPE DEVELOPMENT STAGE**		
009	Frontend MVP development (Dashboards & Visualizer)	Dinesh	14 days
010	Backend server setup, API routing, and DB connection	Yang Junkai	10 days
011	Integration of Generative AI Chatbot & Mentor matching	Aloysius	10 days
012	Basic hashing script for Verified Skills prototype (SHA-256)	Zhang Shaojie (Security)	7 days
Phase 4	**TESTING & CLIENT DEMO STAGE**		
013	Prototype Integration Testing (Linking frontend to backend)	Yang Junkai & Dinesh	5 days
014	Usability Testing (UAT) & Core Bug Fixing	Zhang Zhengyi (QA)	7 days
015	Client Demo Preparation & Final Presentation	All Members	4 days

1.5 **Project Organization**

1.5.1 Project Organization Chart

Provide a graphical depiction of the project team. The representation should be a hierarchical diagram of the project organization, beginning with the project sponsor and including the project team and other stakeholders. Kindly include a photo of each team member, with their roles and responsibilities clearly stated beneath their respective photos.



1.5.2 Roles and Responsibilities

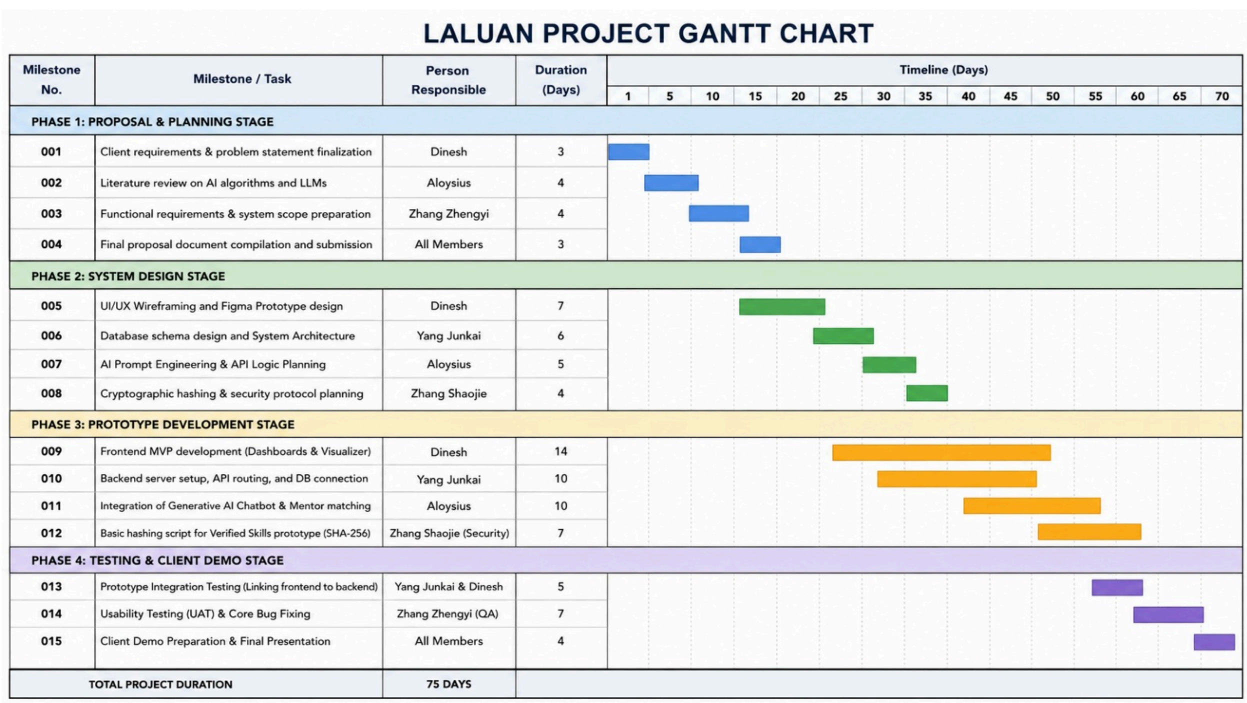
Describe, at a minimum, the roles and responsibilities of all stakeholders identified in the organizational diagram above.

Name	Roles	Responsibilities
Maryam Gamal Ibrahim Abuelgasim	Industry Client	To provide information on the key deliverables of the project and monitor project progress including providing any other assistance deemed important.
Hemalatha Ramalingam	Project Supervisor	To provide guidance and support in term of knowledge, opinion and view throughout the entire project and to evaluating the project and provide recommendation for continuous improvement to meet the project objectives
Dinesh A/L S. Rajandran	Group Leader & Frontend Developer	Coordinates team tasks and milestones to ensure the 3-month deadline is met. Leads the UI/UX design (Figma) and develops the interactive web frontend (Skill Gap Visualizer, Guidance Loop) to ensure a seamless user experience.
YANG JUNKAI	Backend & Database Engineer	Designs the core database architecture (Firebase) to efficiently store user profiles, skill records, and the Campus Calendar events. Develops the backend logic to handle data routing between the frontend and the database.
ZHANG SHAOJIE	Security & Verification Engineer	Handles web security and user authentication protocols. Specifically leads the development of the Cryptographic Hashing system (SHA-256) for the Verified Skills Passport to ensure student certificates are tamper-proof.

Name	Roles	Responsibilities
ZHANG ZHENGYI	QA Tester & System Analyst	Leads the User Acceptance Testing (UAT) phase. Validates the logic behind the Skill Gap analysis, tests the AI responses for accuracy, and ensures the entire application functions flawlessly without bugs before the final demo.
Aloysius Ratnam	AI Engineer	Manages the integration of Generative AI (LLMs) via APIs. Designs the complex system prompts for the AI Career Chatbot and Career Orchestrator to ensure highly accurate, personalized roadmap generation based on student data.

1.6 Gantt Chart

A Gantt chart is a project management tool used to visually represent the timeline of tasks in a project.
Start of week 2 till week 14



LALUAN PROJECT GANTT CHART

Milestone No.	Milestone / Task	Person Responsible	Duration (Days)
PHASE 1: PROPOSAL & PLANNING STAGE			
001	Client requirements & problem statement finalization	Dinesh	3
002	Literature review on AI algorithms and LLMs	Aloysius	4
003	Functional requirements & system scope preparation	Zhang Zhengyi	4
004	Final proposal document compilation and submission	All Members	3
PHASE 2: SYSTEM DESIGN STAGE			
005	UI/UX Wireframing and Figma Prototype design	Dinesh	7
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015	Client Demo Preparation & Final Presentation	All Members	4
TOTAL PROJECT DURATION		75 DAYS	

Chapter 2

H. Appendices

Include any relevant appendices, if applicable. (you may include relevant research papers, or guiding articles that have become the basis of your research/work)

Lee, H.-P., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). *The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers*. Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. ACM.
ACM Digital Library

F1000Research. (2026). *AI-driven career guidance to reduce vocational students' career path anxiety through skills mapping, adaptive mentoring, and labor market intelligence*.
PMC Article

Universiti Kebangsaan Malaysia. (2025). *Graduate employability among graduates in Malaysian public universities*. UKM Institutional Repository.
UKM Repository PDF